Payments to grow your world

Unlocking AI: Navigating Open Source vs. Commercial Frontiers



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Centrul Regional de Afaceri, Timișoara

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I ♥ [Tech]

Who are we?

We design payments technology that powers the growth of millions of businesses around the world.



7000+ engineers
in over 40 countries



Managing 43+ billion
transactions per year



€250M spent in R&D
every year



Handling 150+
payment methods

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The **early days** of LLMs

From rule-based and simpler statistical models to LLMs



2010's

Word embeddings such as Word2Vec and GloVe



2017-2018

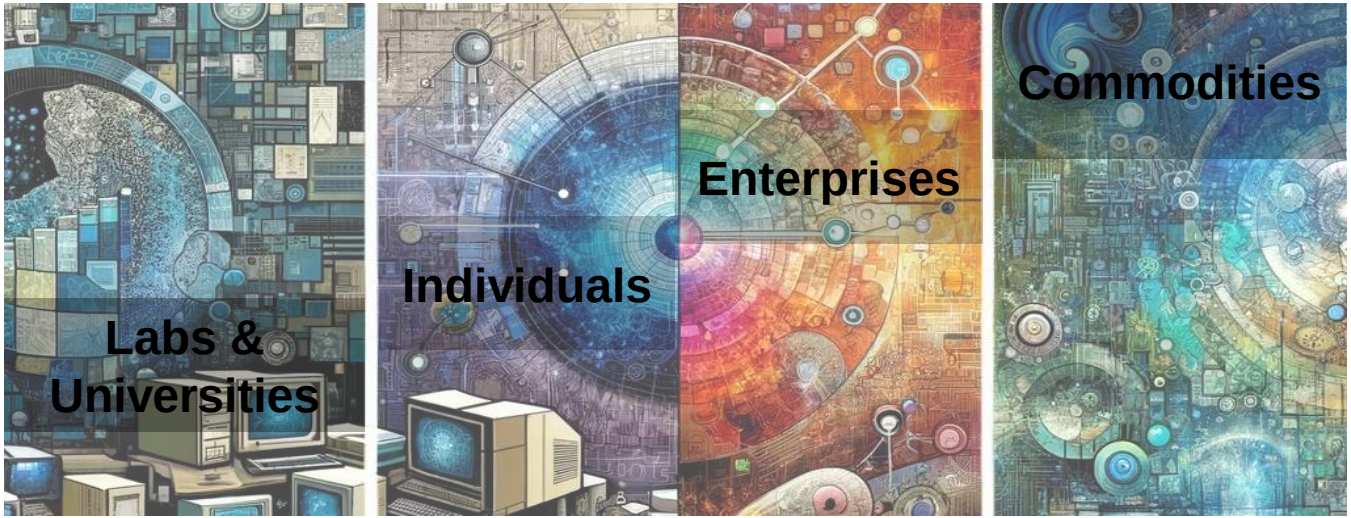
"*Attention is All You Need*" Transformers, BERT



2020's

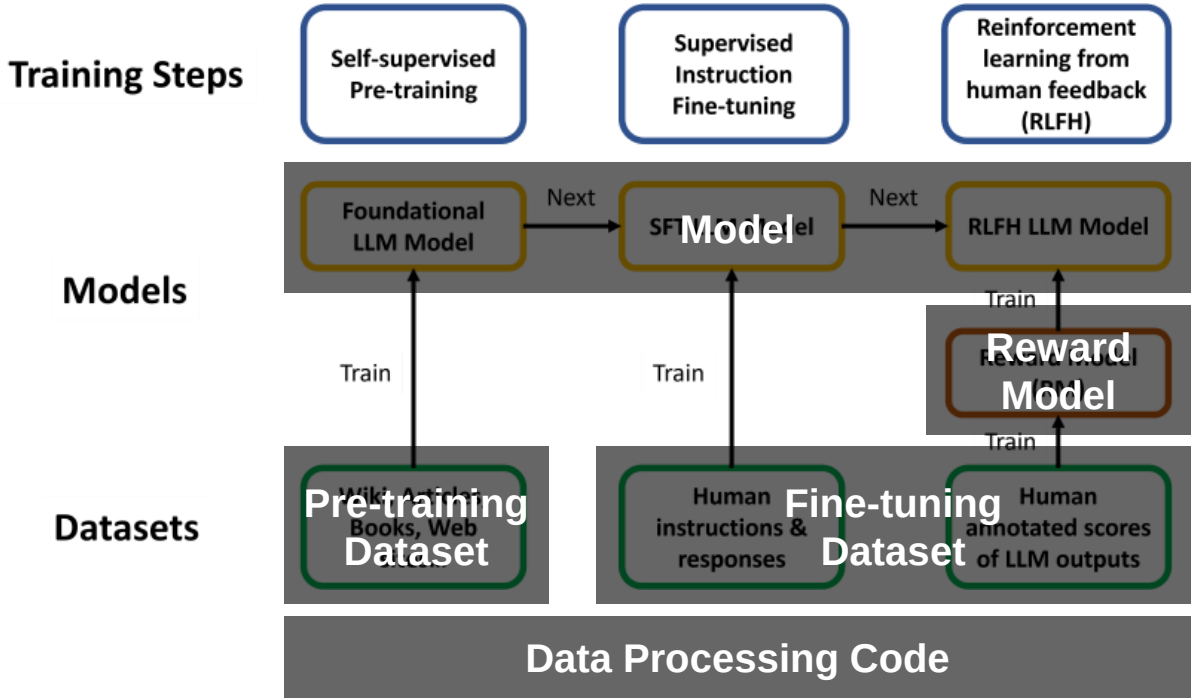
Generative AI, ChatGPT responsibility concerns

GenAI is having its **Linux Moment**



- Just like open source and Internet, bust much faster!
- Dynamics between collaborative openness and commercial ownership
- Need of clarity on licenses

Defining **Openness** of a LLM



Defining **Openness** of a LLM

Score	Level	Description	Model (weights)	Pre-training Dataset	Fine-tuning Dataset	Reward model	Data Processing Code
0	**Closed**	**No access** to any public information, data or asset					
1	**Published research only**	**Research papers(s)** published but with no more information, data or asset					
2	**Restricted access**	Access to asset is possible only with **special agreement** (commercial, research…)					
3	**Open with limitations**	Access and reuse of asset is possible but with certain **limitations on usage**					
4	**Totally open**	Access and reuse of asset is possible **without restriction on usage** (ex. open source license)					

Market-Leading Player: OpenAI

Deviation from original vision of research transparency & openness



GPT-1 & 2

Non/For-profit (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	1	Published research only
Code	1	Published research only

→

GPT-3 & 4

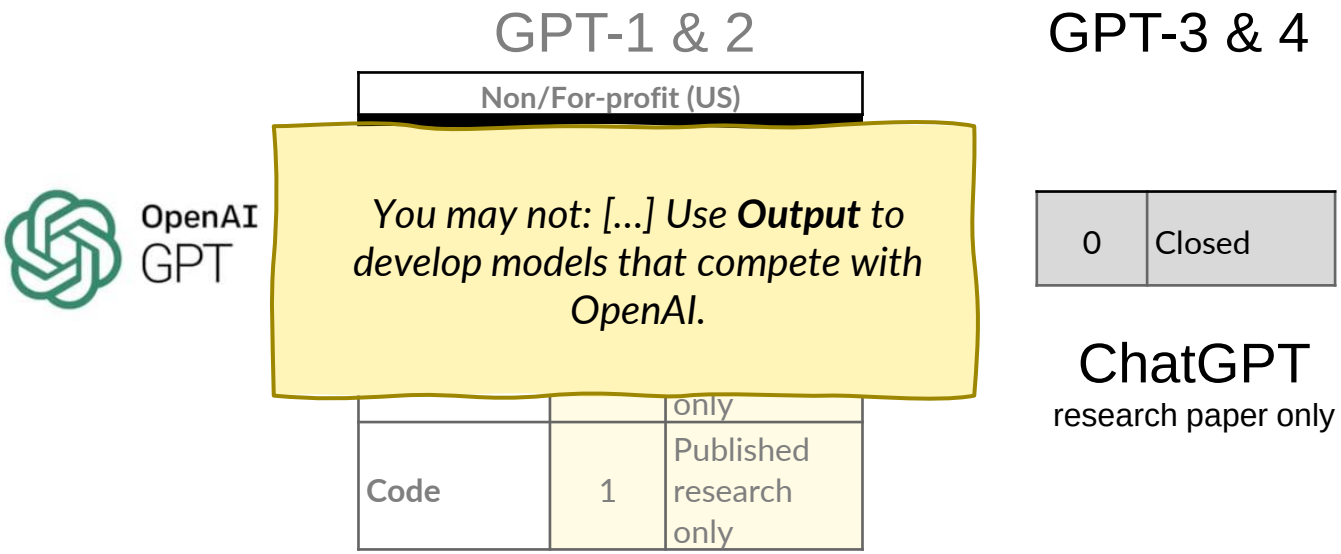
0	Closed

ChatGPT

research paper only

Market-Leading Player: OpenAI

Deviation from original vision of research transparency & openness



No training of other commercial LLMs

Market-Leading Player: Google

Transition from open research to a pragmatic approach

Google BERT

Enterprise (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	2	Restricted access
Code	4	Totally open

→

PaLM 2

Gemini

1	Published research only
1	Published research only
0	Closed

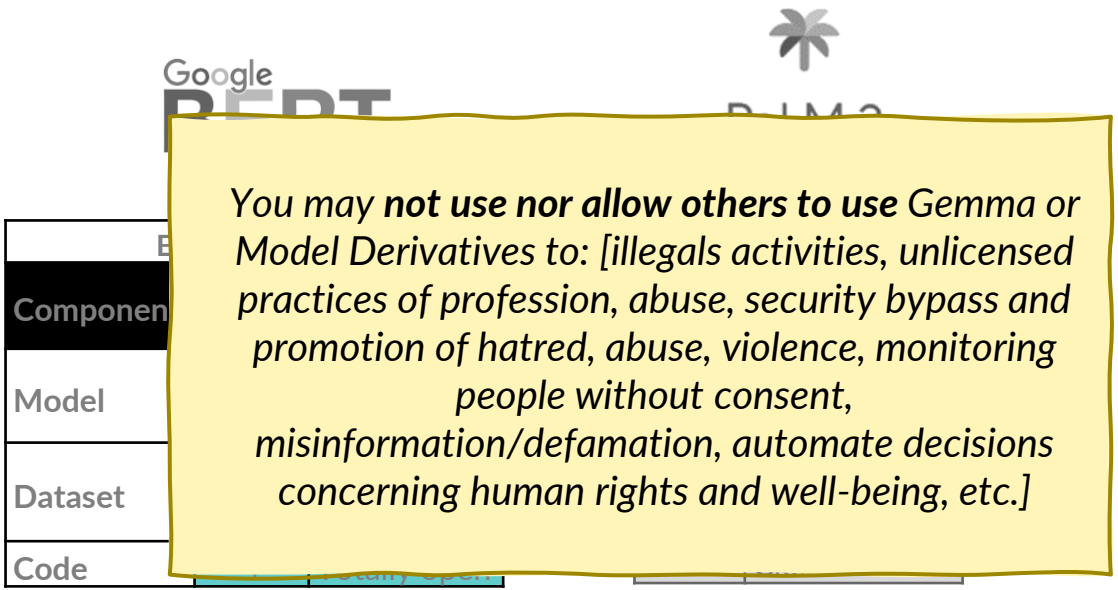
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Gemma

3	Open with limitations
1	Published research only
4	Toolchain available

Market-Leading Player: Google

Transition from open research to a pragmatic approach



*You may **not use nor allow others to use** Gemma or Model Derivatives to: [illegals activities, unlicensed practices of profession, abuse, security bypass and promotion of hatred, abuse, violence, monitoring people without consent, misinformation/defamation, automate decisions concerning human rights and well-being, etc.]*

Gemma

3	Open with limitations
1	Published research only
4	Toolchain available

Responsible AI contradicts Open Source Definition

Market-Leading Player: Meta

Journey to openness

RoBERTa

Enterprise (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	3	Open with limitations
Code	4	Totally open

→

LLaMA by Meta

3	Open with limitations
1	Published research only
1	Published research only

Market-Leading Player: Meta

Journey to openness

RoBERTa

LLaMA

Additional Commercial Terms. *If, on the Llama 2 version release date, the monthly active users of the products or services made available by or for Licensee, or Licensee's affiliates, is greater than 700 million monthly active users in the preceding calendar month, you must request a license from Meta, which Meta may grant to you in its sole discretion, and you are not authorized to exercise any of the rights under this Agreement unless or until Meta otherwise expressly grants you such rights.*



Restriction on usage: license for platforms with 700+ M users

Llama offspring's: Alpaca and Vicuna

Fine-tuned models from Llama 2 by universities





Research (US)		
Component	**Score**	**Level description**
Model	3	Open with limitations
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	2	Research use only
Code	4	Under Apache 2 license



Restrictions from both Llama 2 and OpenAI (ShareGPT)

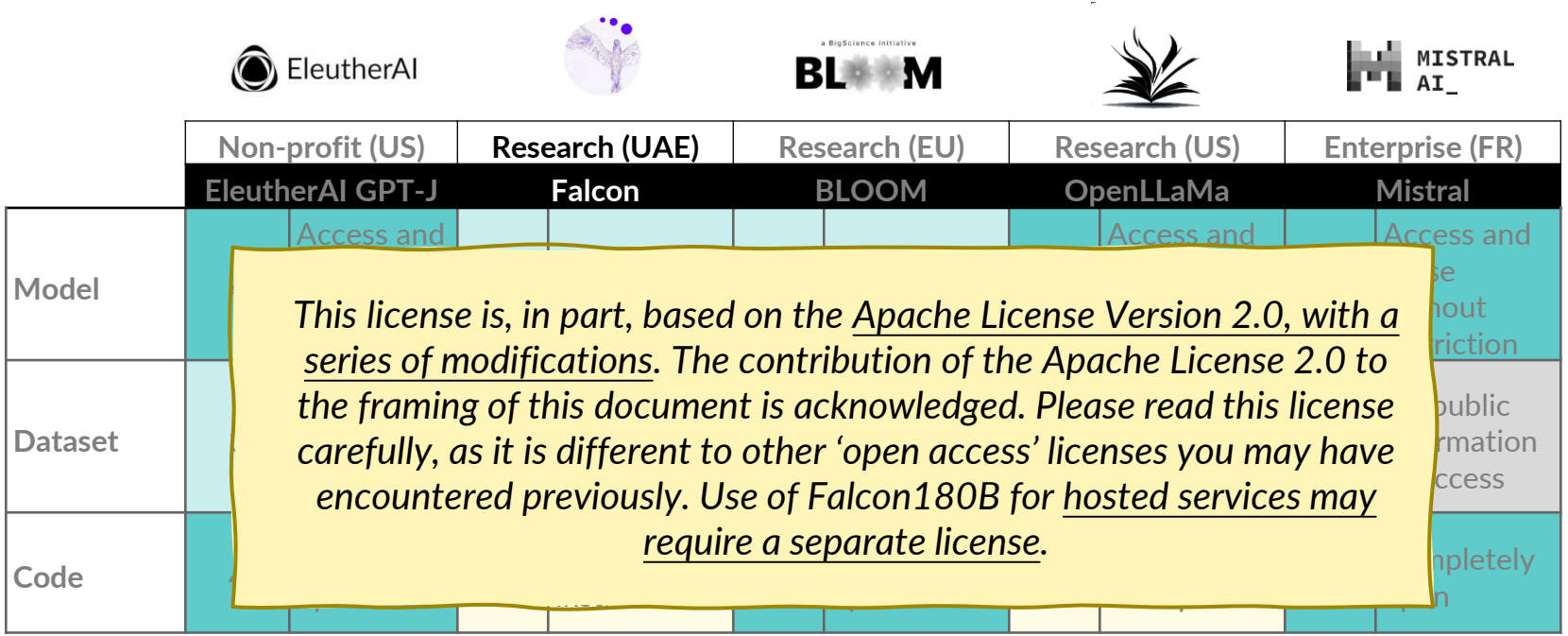
Collaborative **foundational** LLMs

	Non-profit (US)		Research (UAE)		Research (EU)		Research (US)		Enterprise (FR)	
	EleutherAI GPT-J		Falcon		BLOOM		OpenLLaMa		Mistral	
Model	4	Access and reuse without restriction	3	Open with limitations	3	Open RAIL license	4	Access and reuse without restriction	4	Access and reuse without restriction
Dataset	3	Open with limitations	4	Access and reuse without restriction	3	Open with limitations	4	Access and reuse without restriction	0	No public information or access
Code	4	Completely open	1	General instructions	4	Completely open	1	Just examples	4	Completely open

Dataset fuzziness: *please refer to the specific license depending on the subset you use*

Notion of responsible usage

Collaborative **foundational** LLMs



Dataset fuzziness: *please refer to the specific license depending on the subset you use*

Notion of responsible usage

Collaborative **fine-tuned** LLMs

	Enterprise (US)		Enterprise (US)		Enterprise (US)		Consortium (UAE/US)	
	Dolly		BLOOMChat		Zephyr		LLM360	
Model	4	Based on GPT-J	3	Based on BLOOM	4	Based on Mistral	4	Open source
Pre-training Dataset	3	Based on GPT-J	3	Based on BLOOM	0	Based on Mistral	4	RedPajama, Falcon, StarCoder
Fine-tuning Dataset	4	Access and reuse without restriction	4	Dolly and LAION	2	Research use only (OpenAI)	2	Research use only (OpenAI)
Reward model	0	No public information available	0	No public information available	3	Paper and code examples	0	No public information available
Code	4	Open source	3	OpenRAIL	3	Example code available	4	Open source



Impact of foundational model or pre-training datasets

Collaborative **fine-tuned** LLMs

	Dolly		BLOOMChat		Zephyr		LLM360	
	Enterprise (US)		Enterprise (US)		Enterprise (US)		Consortium (UAE/US)	
Model	4	Based on GPT-J	3	Based on BLOOM	4	Based on Mistral	4	Open source
Pre-training Dataset								a, arCoder
Fine-tuning Dataset								use only
Reward model								n
Code	4	Open source	3	OpenRAIL	3	Example code available	4	Open source

BLOOMChat Use Restrictions
l. To provide medical advice and medical results interpretation; or
m. To generate or disseminate information for the purpose to be used for administration of justice, law enforcement, immigration or asylum processes, such as predicting an individual will commit fraud/crime commitment.

Impact of foundational model or pre-training datasets

Collaboration **platform**: Hugging Face

Enabler for collaboration and reuse

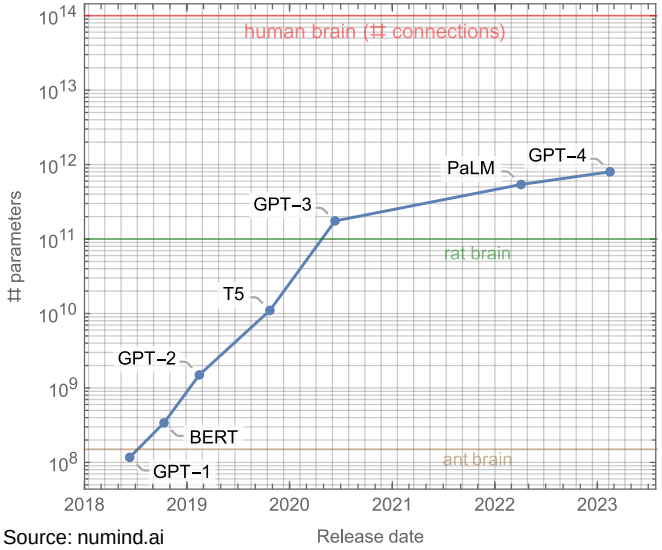


- Startup and ecosystem dedicated to **democratizing AI**
  - Open source Transformers library
  - LLM leaderboard: upload and assess models

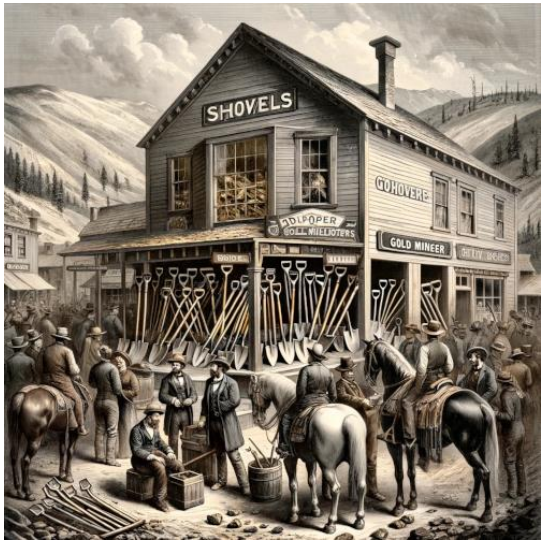
- The "**GitHub of AI**" Models 489,090 Datasets 102,662
  - Collaborative space for exploring, sharing and experimenting AI
  - Hosts thousands of models, datasets, and demo applications

Hosting and resource **paradigms**

Closed models are centralized and resource-consuming



Source: numind.ai



Big players invest billions (Microsoft/OpenAI, AWS/Anthropic)
CSP selling shovels in the AI Gold rush

Hosting and resource **paradigms**







- **Democratizing** AI Computing
  - Quantization, AI Chips
  - Run models locally, in containers
- Emergence of **smaller models** for edge and mobile
  - Small/Tiny Language Models: Gemini nano, Microsoft Phi-2, Huawei TinyBERT
  - Domain Specific Language Models: BloombergGPT, BioMistral, Harvey (law)
  - Mixture of models: Mixtral 8x7B, OpenMoE → Mixture of licenses?

Key **takeaways**



- **Hyper-centralization** leads to black boxes and closed solutions
- **Openness**
  - Fosters collaboration and fuels community-driven innovation
  - Enables inclusivity
- Just like opensource software beware of **licenses and restrictions**
- **GenAI's innovation** continually reshapes the landscape

Thank you



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https://blog.worldline.tech
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Check the two-part article co-written with Luxin Zhang

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